

S.No	Particulars	Theory		Practical	
		Amplifier	Frequency	Amplitude	Time Frequency
1.	Message Signal	1 V	70 Hz	1 V	$T_m = x_2 - x_1$ $= 0.01433$ $f_m = 70 \text{ Hz}$
2.	Carrier Signal	4 V	700 Hz	4 V	$T_m = x_2 - x_1$ $= 0.01433$ $f_c = 700 \text{ Hz}$
3.	AM Wave				$T_m = 0.01433$ $f_{am} = 700 \text{ Hz}$

Ex: No: 2

21/7/26

DOUBLE SIDE BAND SUPPRESSED CARRIER [DSB - SC]

Aim:

To generate double sideband suppressed carrier using SCILAB.

APPARATUS REQUIRED:

- ★ Computer with i3 processor
- ★ SCILAB software.

CODE:

```
clc;
```

```
clear all;
```

```
close;
```

```
Am = 1; Fm = 1500 70;
```

```
Ac = 4; Fc = 150000 700;
```

```
fs = 100000;
```

```
t = 0: 2 / fs; 0.03;
```

```
m = Am * cos(2 * pi * fm * t);
```

```
m = Am * cos(2 * 3.14 * 1500 * t);
```

```
c = Ac * cos(2 * 3.14 * fc * t);
```

```
dsb = m * c;
```

Calculation:

1. Message Signal:

Practical time:

$$\begin{aligned}T_m &= x_2 - x_1 \\&= 0.025 + 0.002 \\&= 0.0143\end{aligned}$$

Practical frequency:

$$\begin{aligned}f_m &= \frac{1}{T_m} \\&= \frac{1}{0.0143}\end{aligned}$$

2. Carrier Signal

Practical time:

$$\begin{aligned}T_c &= x_2 - x_1 \\&= -0.0015 + 0.00293 \\&= 0.001433\end{aligned}$$

Frequency:

$$\begin{aligned}F_c &= \frac{1}{T_c} \\&= \frac{1}{0.001433} \\&= 700 \text{ Hz}\end{aligned}$$

```
subplot (3, 1, 1);
```

```
plot (t, m);
```

```
x title ("Message signal", "Time (s)", "Amplitude");
```

```
subplot (3, 1, 2);
```

```
plot (t, c);
```

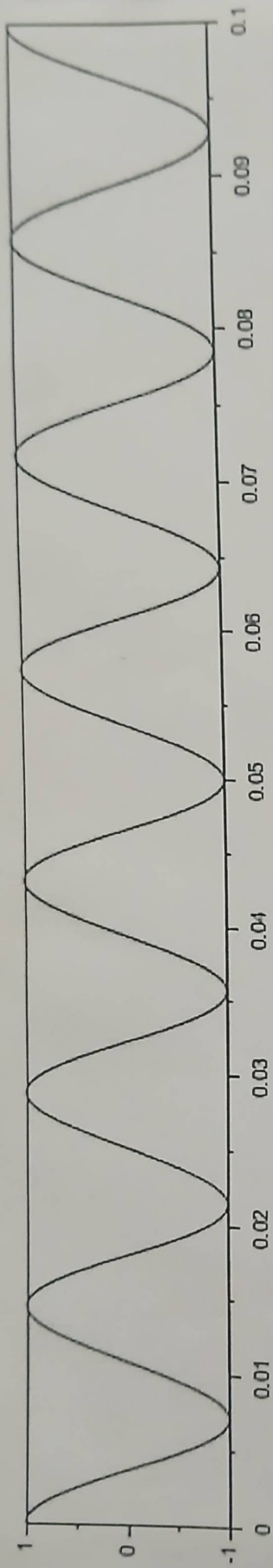
```
x title ("Carrier signal", "Time (s)", "Amplitude");
```

```
subplot (3, 1, 3);
```

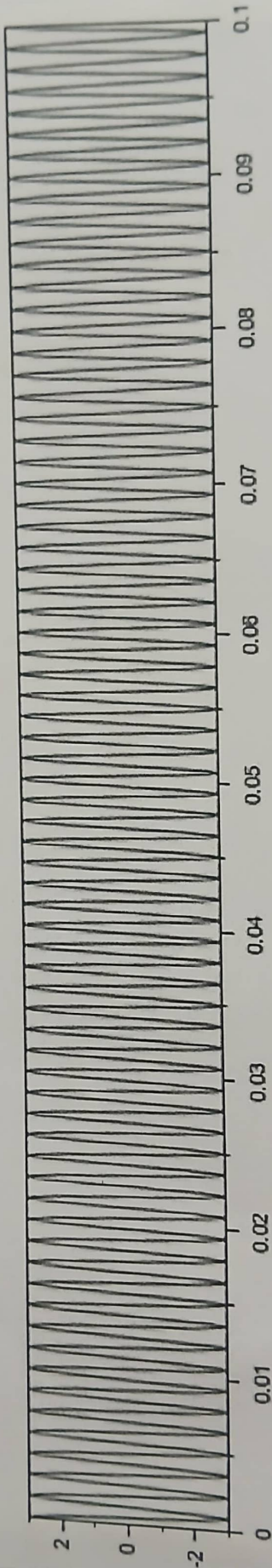
```
plot (t, dsb.sc);
```

```
x title ("DSB "DSB.sc signal", "Time (s)", "Amplitude");
```

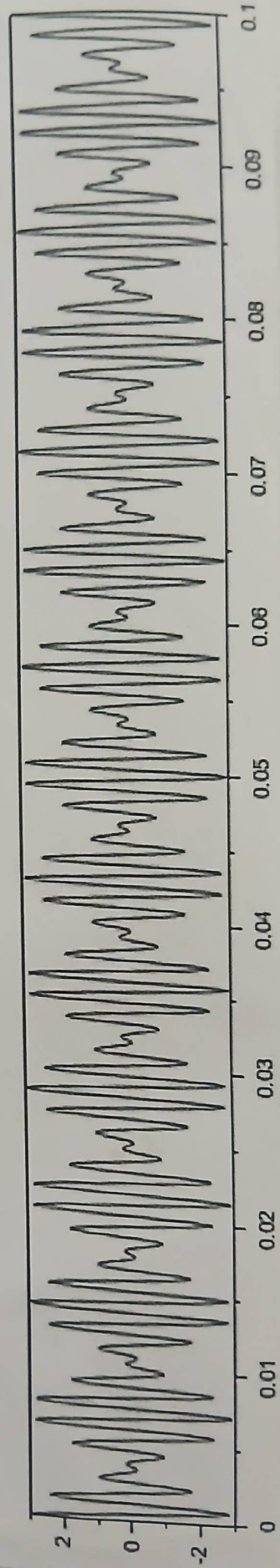
Message Signal



Carrier Signal



DSB-SC Signal



Scilab Implementation (20)	Wave form tabulation (20)	Calculation & Result (20)	Viva voice (20)	Record Result observation (20)	Total mark
20	20	20	15	20	95

RESULT:

Thus the Amplitude modulated (AM) Signal was successfully done using SCILAB.